

The Spike, the Sparse and the Sink: Anatomy of Massive Activations and Attention Sinks

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ONE-LINE TAKEAWAY: The co-occurrence of massive activations and attention sinks in Transformer models is largely an architectural artifact of pre-norm configuration, with massive activations acting globally as implicit parameters and attention sinks modulating attention locally toward short-range dependencies.

PROBLEM

- Massive activations and attention sinks frequently co-occur in modern Transformers, but their functional roles and causal relationship are unclear.
- These phenomena have practical implications for quantization, pruning, KV-cache management, and long-context inference.

METHOD

- Study two phenomena: massive activations (extreme outliers in a few channels) and attention sinks (tokens attracting disproportionate attention).
- Use systematic experiments on Llama and Qwen models to analyze activation patterns across layers.
- Identify the role of pre-norm configuration in enabling co-occurrence.
- Ablate normalization to decouple the phenomena.
- Analyze how massive activations propagate through residual connections and influence attention.

KEY NUMBERS

64	Number of blocks in Llama 2 7B model.
72	Number of blocks in Qwen3 8B model.
10	Step-up block index in QWEN2.5 14B model.
95	Step-down block index in QWEN2.5 14B model.



CONTRIBUTION

A principled bisection of massive activations and attention sinks, revealing their architectural roots, mechanisms, and independent controllability.



RESULTS



Massive activations appear only in intermediate layers, affect a small number of channels, and maintain fixed inter-channel ratios.



Step-up blocks introduce massive activations early; step-down blocks neutralize them near the end.



Abolishing pre-norm causes massive activations and attention sinks to decouple.



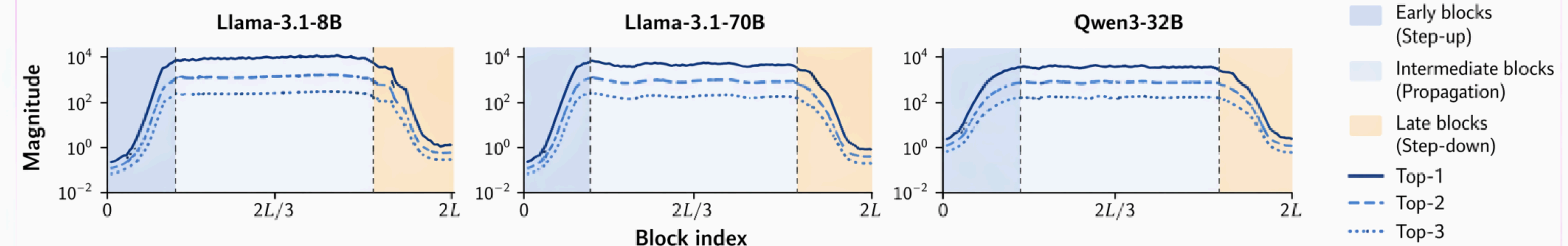
Attention sinks are driven by attention space dimensionality and training context-length distribution.



Each phenomenon can be independently suppressed without degrading language-modeling performance.

Figure 1: The life cycle of massive activations. Top panels show the magnitude of

Top-3 channel magnitudes of post-residual hidden representations



Top-3 channel magnitudes of individual block outputs (spike channels)

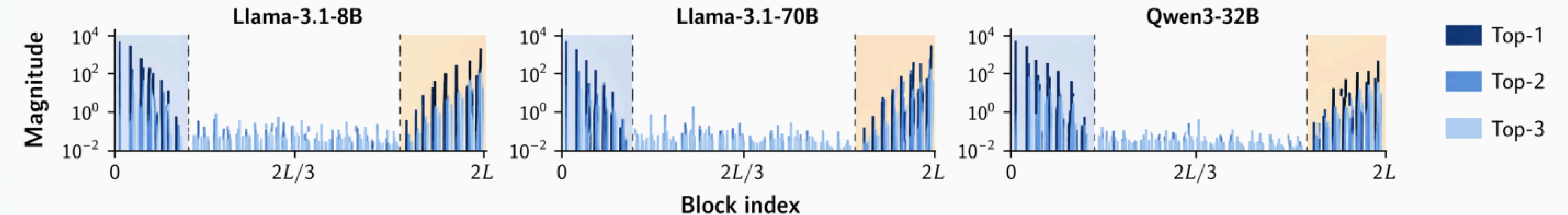


Figure 3. Frobenius norms $\|U_k\|_F$ for the quadratic forms in Llama 2 7B. Spike

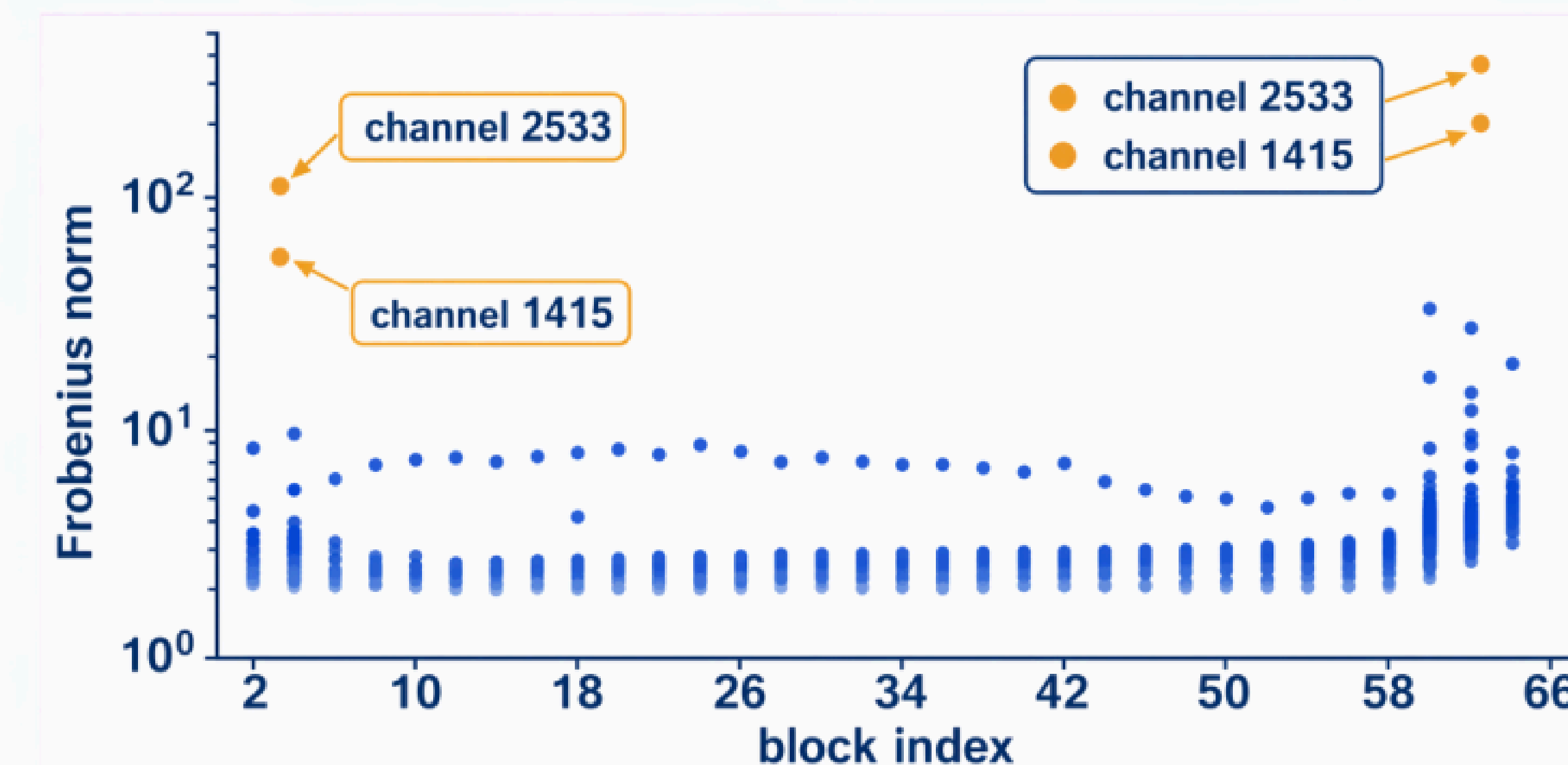
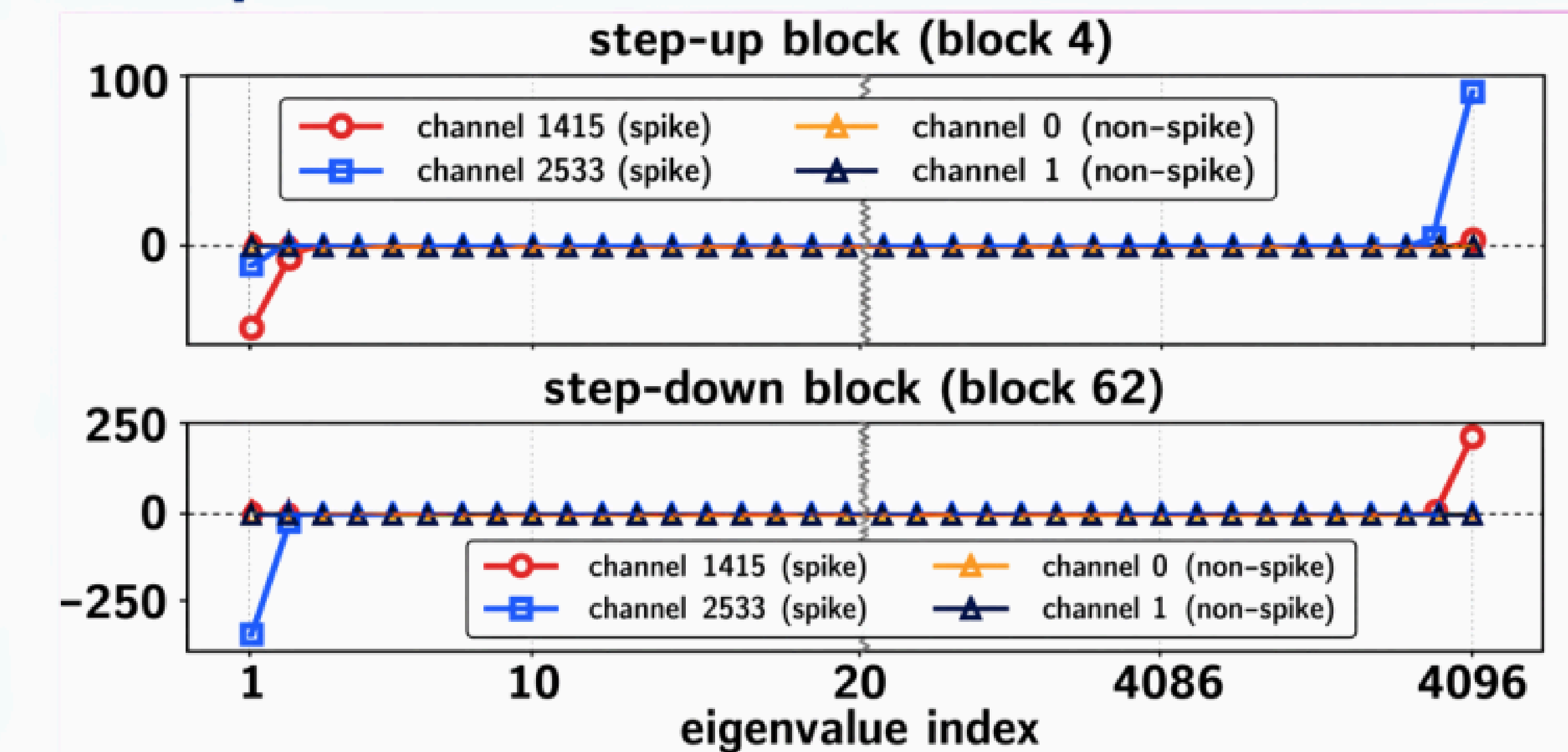


Figure 4. Eigenvalue spectra of S_k for spike vs. non-spike channels in Llama 2



IMPLICATIONS

- Improved quantization and pruning by identifying critical channels.
- Better KV-cache management for long-context inference.
- Guidance for architecture design beyond pre-norm configurations.



EXPERIMENTAL SETTING

Systematic experiments on Llama and Qwen models, analyzing activation patterns across layers and ablating normalization.